

Catalytic competence bounds the decay rates of a thermally formed radical pair

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A flavoenzyme could respond to the geomagnetic field through the radical-pair mechanism only if its catalytic cycle passes through a spin-correlated radical pair that lives long enough for a 50 μT field to act. For a pair that is formed thermally from a closed-shell enzyme–substrate complex, returns to it only through the singlet, and carries the catalytic flux, we show that detailed balance bounds both decay rates from below: the singlet back-transfer rate and the spin-independent onward rate must each exceed $k_{\text{cat}}e^{\Delta/k_B T}$, where Δ is the free energy of the pair. The bound rests on an inequality for the onward yield, $\Phi_P \leq 4k_P/k_S$, that holds for any spin Hamiltonian and any unital, positivity-preserving spin relaxation, and it involves no electronic coupling or reorganization energy. We combine it with a numerical search for the largest field effect at given rates in a flavin–aminium spin model with density-functional hyperfine and dipolar couplings. Among the searched pairs able to carry $k_{\text{cat}} = 1 \text{ s}^{-1}$, the largest effect found falls below 0.1 % for $\Delta \geq 0.47 \text{ eV}$, or 0.50 eV when every searched value is multiplied by the largest factor found with richer nuclear models; electron spin relaxation, computed directly, lowers the values that set this threshold. With a nuclear-frequency ceiling on elementary rates, no pair above 0.86 eV carries that turnover. The measured flavin potentials of the monoamine oxidases, combined with a dialkylamine potential standing in for benzylamine-type substrates, place an amine-to-flavin pair 1.41–1.44 eV above the closed-shell state, and $0.91 \pm 0.05 \text{ eV}$ if substrate binding shifted the one-electron flavin couple as much as it shifts full reduction.

I. INTRODUCTION

The radical-pair mechanism (RPM) is the only experimentally supported way in which a magnetic field as weak as the geomagnetic field ($\sim 50 \mu\text{T}$) can alter the yield of a chemical reaction.^{1,2} It underlies the leading model of avian magnetoreception, in which photoexcitation of the flavin in cryptochrome creates a spin-correlated flavin–tryptophan radical pair.^{3–5} Whether the same mechanism could act in flavoenzymes that work in the dark, such as the monoamine oxidases that degrade neurotransmitters, has been raised repeatedly.^{6,7}

General limits on weak-field effects are known. Electron spin relaxation caps a primary RPM effect in biochemistry at 10^{-3} – 10^{-4} or less,⁸ and exchange and dipolar couplings between the radicals suppress singlet–triplet interconversion unless the radicals are far apart.⁹ Asymmetric reaction schemes, in which only the singlet pair recombines while both spin states react onward, relax the second limit and allow closely bound pairs to respond.¹⁰ These results constrain the spin dynamics of a pair with given lifetimes. They do not say which lifetimes a pair that an enzyme forms *thermally* can have.

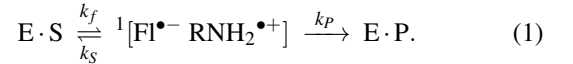
Here we derive that constraint. For a pair formed from a closed-shell complex that must carry the catalytic flux, detailed balance bounds both of its decay rates from below in terms of the driving force Δ and the turnover number. The bound is the spin-resolved form of the familiar limit on flux through a pre-equilibrium intermediate; what the spin dynamics adds is that it holds for the spin-independent onward rate as well as for the singlet return, with a factor of four that is sharp for any spin Hamiltonian. We then combine it with a numerical search for the largest field effect at given rates in a flavin–aminium spin model, state how that search depends on the model, and estimate where the monoamine oxidase pair

lies from the measured flavin potentials of the enzyme and a proxy for the amine potential.

II. REACTION SCHEME AND SPIN MODEL

A. Scheme and observable

We consider the single-electron-transfer (SET) scheme proposed for amine oxidation by flavoenzymes:¹¹



Electron transfer from the closed-shell complex forms the pair in its singlet state; only the singlet pair returns by back electron transfer (rate k_S), and singlet and triplet pairs react onward with a spin-independent rate k_P , for example by $\alpha\text{-C-H}$ deprotonation of the aminium radical. The yield of onward reaction per pair formed is $\Phi_P = k_P \int_0^\infty \text{Tr} \rho \, dt$, and the magnetic field effect is $\text{MFE}_P = [\Phi_P(B) - \Phi_P(0)]/\Phi_P(0)$ at $B = 50 \mu\text{T}$. The spin density operator of a pair born in the singlet evolves as

$$\dot{\rho} = \mathcal{L}\rho = -2\pi i[H, \rho] - \frac{k_S}{2}\{P_S, \rho\} - k_P\rho + \mathcal{D}\rho, \quad (2)$$

$$\rho(0) = P_S/N,$$

with P_S the singlet projector, $N = \text{Tr} P_S$ the nuclear-spin dimension and $\mathcal{D}$ a spin-relaxation superoperator. We assume that $\mathcal{D}$ generates a positivity-preserving evolution, as a trace-preserving Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) generator does, and that it is unital, $\mathcal{D}(I) = 0$, as single-electron dephasing, the reactive singlet–triplet dephasing of ref 12 and any relaxation that drives the spins towards

equal populations at high temperature are. Yields are obtained exactly, from the Sylvester equation for $\int \rho dt$ in the coherent case and in Liouville space with relaxation.

B. Spin Hamiltonian and conventions

In frequency units,

$$H = \omega_B \hat{n}_B \cdot (\mathbf{S}_1 + \mathbf{S}_2) + \sum_k \mathbf{S}_{e(k)} \cdot \mathbf{A}_k \cdot \mathbf{I}_k + J \mathbf{S}_1 \cdot \mathbf{S}_2 + \mathbf{S}_1 \cdot \mathbf{T} \cdot \mathbf{S}_2, \quad (3)$$

with $\omega_B = g\mu_B B/h$ and hyperfine tensors $\mathbf{A}_k$. $J = E_T - E_S$ is the full singlet-triplet gap; in the convention of Efimova and Hore⁹ $J = -2J_{\text{EH}}$. $\mathbf{T}$ is the traceless dipolar tensor; for an axial tensor $\mathbf{S}_1 \cdot \mathbf{T} \cdot \mathbf{S}_2 = 2D[(\mathbf{S}_1 \cdot \hat{n})(\mathbf{S}_2 \cdot \hat{n}) - \mathbf{S}_1 \cdot \mathbf{S}_2/3]$, so that with the field along $\hat{n}$ the $T_{\pm}$ and T_0 levels are split by D ($D = -77.9$ MHz for point dipoles 1 nm apart⁹). Both conventions are verified numerically in the code.

C. Magnetic parameters

The reference model has four nuclei: flavin N5 (523 μT) and N10 (189 μT), each as an isotropic spin-1 nucleus,¹³ and the two largest couplings of the benzylaminium radical cation, 257 and 196 MHz ($\alpha\text{-CH}_2$ protons; B3LYP/def2-TZVP on GFN2-xTB geometries¹⁴). The dipolar tensor of the contact pair is the point-dipole tensor summed over the Löwdin spin populations of the two radicals for methylamine stacked on the flavin si face at an N5-N distance of 3.5 Å, with $D = -843$ MHz and $E = -26$ MHz. Because the pair geometry is not known, the search below scales this tensor by any factor $s \in [0, 3]$, from no dipolar coupling to three times the contact value. Richer nuclear models and spin relaxation are examined in Sec. VI.

III. CATALYTIC COMPETENCE BOUNDS BOTH DECAY RATES

Let Δ be the free energy of the radical pair relative to E·S, summed over its four spin states, so that, with spin energies negligible against $k_B T$, the singlet pair lies $\Delta_S = \Delta + k_B T \ln 4$ above E·S ($k_B T \ln 4 = 0.037$ eV at 310 K) and detailed balance for the singlet channel gives $k_f = k_S e^{-\Delta_S/k_B T}$. The turnover carried by the pair is $k_f \Phi_P$ per E·S complex, and two inequalities bound Φ_P . Trivially $\Phi_P \leq 1$. Less obviously,

$$\Phi_P \leq 4k_P/k_S \quad (4)$$

for any spin Hamiltonian and any $\mathcal{D}$ with the properties above. Indeed, for $k_P > 0$ the resolvent $\mathcal{R} = (-\mathcal{L})^{-1} = \int_0^\infty e^{\mathcal{L}t} dt$ exists and maps positive operators to positive operators, and unitality gives $-\mathcal{L}(I) = k_S P_S + k_P I$; applying $\mathcal{R}$, $k_S \mathcal{R}(P_S) = I - k_P \mathcal{R}(I) \preceq I$, and taking the trace, $\Phi_P = (k_P/N) \text{Tr} \mathcal{R}(P_S) \leq 4k_P/k_S$. Over 300 random spin systems with exchange, dipolar and hyperfine couplings and, in part, electron spin relaxation, the largest value of $\Phi_P k_S/4k_P$ is 0.9986 (Supporting Information).

Requiring the turnover $k_f \Phi_P$ to be at least k_{cat} gives

$$k_S \geq k_{\text{cat}} e^{\Delta/k_B T} \quad \text{and} \quad k_P \geq k_{\text{cat}} e^{\Delta/k_B T} \equiv k_{\text{min}}(\Delta). \quad (5)$$

(The first inequality holds with an extra factor of four; we use the weaker form.) Eq 5 is a statement about the generator of eq 2; whether a strongly coupled contact pair is described by such a generator is a separate question, taken up in Sec. VII. Without spin dynamics the turnover is exactly $e^{-\Delta_S/k_B T}/(1/k_S + 1/k_P)$. Eq 5 contains no electronic coupling, no reorganization energy and no assumption about the chemistry of the onward step. If, in addition, no elementary rate exceeds the frequency of nuclear motion, $\nu_n \approx 10^{13} - 10^{14} \text{ s}^{-1}$, a pair that carries $k_{\text{cat}} = 1 - 10^3 \text{ s}^{-1}$ must have $\Delta \leq k_B T \ln(\nu_n/k_{\text{cat}}) = 0.62 - 0.86 \text{ eV}$.

A radical pair on a dead-end branch, which does not lead to product, is a separate case. It carries no flux, so eq 5 does not apply to it, but it holds enzyme, and a field that changes its population changes the free enzyme. With $k_P = 0$ the resolvent need not exist (triplet states that never reach the singlet are stationary), but the identity $k_S \int_0^t e^{\mathcal{L}_0 u} (P_S) du = I - e^{\mathcal{L}_0 t} (I) \preceq I$, with $\mathcal{L}_0$ the generator at $k_P = 0$, bounds the integrated singlet-born population by $4N/k_S$ for all t . The steady-state population of the branch is therefore at most $4k_f/k_S = e^{-\Delta/k_B T}$ per E·S complex, and so is the fraction of enzyme through which a field can act on it.

IV. THE LARGEST FIELD EFFECT FOUND AT GIVEN RATES

The exchange coupling of a contact pair is the sum of a reactive contribution from the electron transfer that sets k_S ¹² and a direct contribution that the rates do not fix. We therefore treat J as free and search, for given (k_S, k_P) , for

$$F^*(k_S, k_P) = \sup_{J, \hat{n}_B, s \in [0, 3]} |\text{MFE}_P|. \quad (6)$$

We write $F \leq F^*$ for the largest value the search finds. The search scans J on a grid that is dense near zero and extends to $\max(4000 \text{ MHz}, 5(k_S + k_P)/2\pi)$, for $s = 0, 0.3, 1$ and 3 and 25 field directions over a full hemisphere, refines each of the three largest local maxima in J , and then polishes the four best candidates by a simplex search in $(J, \hat{n}_B, s)$. Because the response has narrow resonances in s where sD is comparable with the hyperfine and Zeeman terms, a second pass scans s on a fine logarithmic grid (41 values) at a coarse J grid and polishes again. F is the larger of the two passes. The second pass raised 151 of the 195 cells, by up to a factor of 3.6, but changed no row maximum by more than 0.0%; tripling the density of the s grid at the cells that set the thresholds, and at the cell the second pass raised most, changed F by a factor of at most 1.018. No maximum lies at the edge of the J range (0 of 195 cells).

Figure 1 shows F over $k_S = 10^{-2} - 10^5 \mu\text{s}^{-1}$ and $k_P = 10^{-2} - 10^4 \mu\text{s}^{-1}$. The onward rate controls the response: $\max_{k_S} F$ is 14 % at $k_P = 10^{-2} \mu\text{s}^{-1}$, 0.023 % at $10^2 \mu\text{s}^{-1}$ and $1.2 \times 10^{-7} \%$ at $10^4 \mu\text{s}^{-1}$. A fast singlet back-transfer does not by

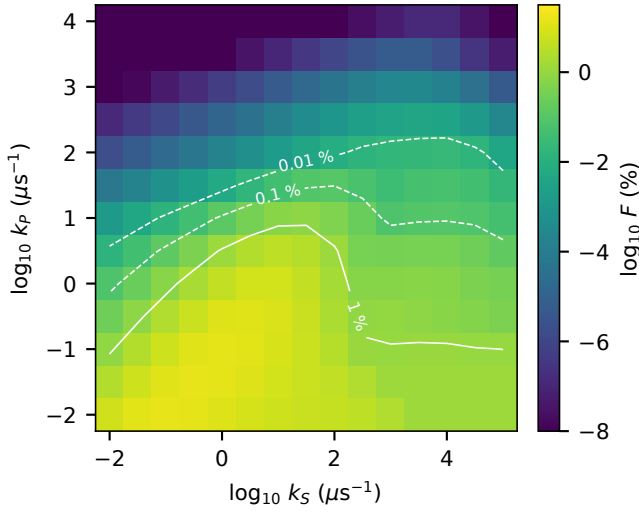


FIG. 1. Largest magnetic field effect on the onward yield found by the search over the exchange coupling, the field direction and the dipolar scale $s \in [0, 3]$, as a function of the singlet back-transfer rate k_S and the spin-independent onward rate k_P . Reference four-nucleus model, coherent spin dynamics, $B = 50 \mu\text{T}$.

itself remove the effect, because the triplet pair cannot return and is drained only by k_P ,¹⁰ but a fast spin-independent decay removes it for every k_S , J and s searched.

V. FIELD EFFECTS OF A COMPETENT PAIR

Combining eq 5 with the field effect defines

$$G^*(\Delta) = \sup\{F^*(k_S, k_P) : k_S, k_P \geq k_{\min}(\Delta)\}, \quad (7)$$

the largest field effect of a pair that can carry the turnover. We evaluate it by a procedure on the rate grid and call the result G . It is not a bound on G^* : between grid points the effect can exceed it. At $\Delta = 0.46 \text{ eV}$, where $k_{\min}$ coincides with the grid rate $k_P = 31.6 \mu\text{s}^{-1}$, the permitted point $k_S = 76 \mu\text{s}^{-1}$ gives 0.0982 % against $G = 0.0965 \%$ (Supporting Information). In the procedure, $k_{\min}$ is rounded down to the nearest grid rate; above the largest k_P the edge value is used, because $\max_{k_S} F$ decreases with k_P on the grid, and above the largest k_S only the constraint on k_P is imposed. Because the permitted region only shrinks as Δ grows, G is finally replaced by its running maximum from larger Δ , which removes non-monotonic steps introduced by these grid rules. Where $k_{\min}$ lies below the grid ($\Delta < 0.25 \text{ eV}$ for $k_{\text{cat}} = 1 \text{ s}^{-1}$) the grid does not cover the permitted rates, and G is not reported; there slower pairs are permitted and give larger effects.

Figure 2 shows G for $k_{\text{cat}} = 1\text{--}10^3 \text{ s}^{-1}$. Because k_P is sampled every half decade, the thresholds below are resolved to about 0.03 eV. For $k_{\text{cat}} = 1 \text{ s}^{-1}$, G falls below 0.1 % for $\Delta \geq 0.47 \text{ eV}$ and below 0.01 % for $\Delta \geq 0.53 \text{ eV}$; for $k_{\text{cat}} = 10^3 \text{ s}^{-1}$ the 0.1 % threshold is 0.28 eV. A pair near thermoneutrality is not excluded: at small Δ competence permits slow decay and the field effect can be several percent.

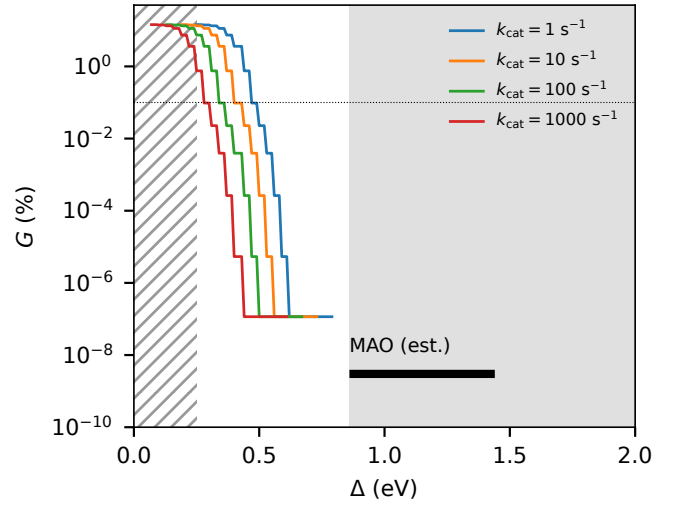


FIG. 2. Largest magnetic field effect found by the grid procedure among pairs that can carry the turnover, $G(\Delta)$, for $k_{\text{cat}} = 1, 10, 100$ and 10^3 s^{-1} in the reference model. Hatched: G not defined for $k_{\text{cat}} = 1 \text{ s}^{-1}$ because the permitted rates extend below the computed grid. Shaded: no pair carries the turnover with $v_n = 10^{14} \text{ s}^{-1}$. Bar: driving force of an amine-to-flavin pair in monoamine oxidase estimated from measured flavin potentials and a proxy amine potential (Sec. VII).

VI. DEPENDENCE ON THE SPIN MODEL

F and G belong to the reference model, and they change with it. We re-searched the cell that sets the row maximum of F for each onward rate that determines the thresholds ($k_P = 10\text{--}10^{2.5} \mu\text{s}^{-1}$) in three richer nuclear models: anisotropic hyperfine tensors for N5 and N10 (density-functional spin-dipolar parts in the frame of the dipolar tensor), the same with the aminium ^{14}N added, and the same with the two largest flavin methyl protons added. Relative to the reference model the largest factors found are 3.1 (anisotropic tensors), 1.6 (with ^{14}N) and 2.2 (with the methyl protons). As a stress test we multiply every searched value by the largest of these, 3.1; this shifts the 0.1 % threshold for $k_{\text{cat}} = 1 \text{ s}^{-1}$ from 0.47 to 0.50 eV. Only the reference row-maximum cell was searched again, so in a richer model another cell of the row could be larger, and the shifted threshold is the result of a stress test, not an upper bound over spin models.

Electron spin relaxation was computed directly in the four-nucleus model. Every cell that the thresholds use on these rows ($k_S \geq k_P$, 90 searches) was searched again with single-electron dephasing, $T_2^e = 10 \text{ ns}$, 100 ns and 1 μs , using a solver that does not build the Liouvillian (Supporting Information). Relaxation changed individual cells by factors of 0.11–4.94, so it does not reduce F everywhere, but it lowered every row maximum: the largest value found with relaxation in each row is 0.84–0.99 times the coherent one. With each cell replaced by the larger of its coherent and relaxed values, the 0.1 % threshold for $k_{\text{cat}} = 1 \text{ s}^{-1}$ is 0.47 eV and the 0.01 % threshold 0.53 eV.

VII. WHERE MONOAMINE OXIDASE LIES

The potentials of the covalent flavin in the monoamine oxidases have been measured. Chemical reduction proceeds in two steps, the first giving the semiquinone and each requiring two electron equivalents of the enzyme; the oxidized/semiquinone midpoint potential is -159 mV in MAO A and -167 mV in MAO B,¹⁵ and the flavin radical of MAO A is anionic,¹⁶ the radical of the scheme above. We take these midpoints as the one-electron flavin couple. For the amine, dialkylamine radical cations in water have $E^\circ(\text{R}_2\text{NH}^{+\bullet}/\text{R}_2\text{NH}) = 1.30 \pm 0.05$ V vs NHE, a value its authors give as a general approximation;¹⁷ we are not aware of a measured potential for a primary alkylamine or benzylamine radical cation in water and use this value as a proxy. With the Coulomb attraction of the contact pair in water (0.03 – 0.05 eV for 6 – 3.5 Å) this places the pair at $\Delta \approx 1.41$ – 1.44 eV.

Substrate binding changes the flavin potential: in the presence of substrates the midpoint potential for full reduction shifts by up to $+0.5$ V.¹⁵ As a bounding assumption, if the one-electron couple shifted by the same amount, the pair would lie at $\Delta \approx 0.91$ eV, or 0.86 – 0.96 eV within the uncertainty of the amine proxy. This range straddles the competence limit of 0.86 eV, so on this assumption the thermodynamics alone does not exclude the pair from carrying turnover. It lies, however, well above the driving forces at which the searched field effect is appreciable ($G < 0.1$ % for $\Delta \geq 0.47$ – 0.50 eV): a pair at the 0.1 % threshold would require the active site to lower Δ by a further 0.41 eV beyond that shift. The kinetic evidence points away from a long-lived radical pair: no flavin radical is detected during the reductive half-reaction of MAO B,¹⁸ and the substituent and isotope effects of MAO A indicate rate-limiting α -C–H cleavage by proton abstraction.¹⁹ These estimates concern substrates of the benzylamine type, oxidized at nitrogen. Substrates that may be oxidized at the ring rather than at nitrogen, such as the catechol and indole neurotransmitters, are not analysed here and would require their own potentials.

The electronic coupling between flavin and substrate does not enter eq 5, but it indicates the regime. Fragment-orbital couplings for methylamine at van der Waals contact (69 non-clashing geometries of 112 ; Supporting Information) are 53 meV (median) to 106 meV at 3.5 Å, large enough that back electron transfer at contact would be adiabatic. The reaction terms of eq 2 are derived for weak, non-adiabatic coupling.¹² Eq 5 holds for any generator of that form with unital, positivity-preserving relaxation, including the reactive dephasing that appears at higher order; whether a strongly coupled contact pair is described by such a generator at all is not addressed here, and F is least reliable for it.

VIII. A LIGHT-DRIVEN PAIR IN THE SAME MODEL

The flavin–tryptophan pair of cryptochrome is formed by light, so eq 5 does not apply to it. As a check that the same equations produce a field response when the rates are not constrained, we evaluated it with the same scheme and

yield observable, averaged over field orientations rather than maximized, $k_S = k_P = 1 \mu\text{s}^{-1}$, the pair 1.90 nm apart ($D = -11.4$ MHz), flavin N5 and N10 as above, the two largest β -proton couplings (60 and 56 MHz) and N1 of the tryptophan radical cation from the same density-functional method, and the exchange coupling on 21 values spanning the literature range of its prefactor with both signs.⁹ The orientation-averaged response ranges from below 10^{-2} % (it changes sign within the range) to 0.63 %. This is a numerical control, not a like-for-like comparison: the hyperfine couplings, dipolar coupling and rates differ from the enzyme model as well as the way the pair is formed.

IX. DISCUSSION

The bound of eq 5 applies to any radical pair that is formed thermally from a closed-shell state, returns to it only through the singlet, and carries the catalytic flux; a pair on a dead-end branch is limited instead by the population bound of the same section. Both require the reaction terms of eq 2 and unital, positivity-preserving relaxation, but no electronic coupling, reorganization energy or particular onward chemistry. The numbers that follow depend on more: F and G are the results of a finite search in a specific spin model with coherent dynamics, the shifted threshold is a stress test, and the monoamine oxidase estimate depends on a dialkylamine potential standing in for the substrates and on how substrate binding shifts the one-electron flavin couple.

Two routes around the bound should be named. Three-radical and scavenging mechanisms²⁰ restore sensitivity to strongly coupled pairs; they still require the pair to survive until the third radical acts, which eq 5 limits, but their field effects would have to be computed in their own model. And an active site that brings Δ close to zero is not excluded: competence then allows slow decay and field effects of a percent or more. The question for a particular enzyme is where its radical pair lies in Δ , which is measurable.

X. CONCLUSIONS

For a radical pair that an enzyme forms thermally and through which it turns over, detailed balance bounds the singlet back-transfer rate and the spin-independent onward rate from below by $k_{\text{cat}} e^{\Delta/k_B T}$, for any spin Hamiltonian and unital, positivity-preserving relaxation. In a flavin–aminium spin model with computed hyperfine and dipolar couplings, the largest field effect found among the searched pairs that can carry $k_{\text{cat}} = 1 \text{ s}^{-1}$ falls below 0.1 % for $\Delta \geq 0.47$ eV (0.50 eV in a stress test for richer nuclear models, and 0.47 eV with electron spin relaxation included). The measured flavin potentials of the monoamine oxidases, with a dialkylamine proxy for the potential of benzylamine-type substrates, place an amine-to-flavin pair at 0.86 – 1.44 eV, depending on how substrate binding shifts the one-electron couple, above that range.

SUPPLEMENTARY MATERIAL

See the supplementary material for the 300-system test of eq 4, the full search results, the electron-spin-relaxation calculations in the four-nucleus model, the hyperfine and dipolar tensors, the coupling scan, the density-functional driving forces, the model-sensitivity calculations, the cryptochrome scan, and software and environment.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Hikaru Wakaura: Conceptualization (lead); Methodology (lead); Software (lead); Validation (lead); Formal analysis (lead); Investigation (lead); Data curation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (lead). **Taiki Tanimae:** Supervision (lead).

DATA AVAILABILITY

The data that support the findings of this study and the code that generates them are openly available in the qbscreen

package at https://github.com/deeptell-inc/brain_protein_screening; every number in the text is generated from the stored results by `qbscreen/make_numbers_v6.py`.

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